

Castalia

A four-channel electrochemical sensing SoC with a five-hart RISC-V back end

Maxx Seminario

University of Nebraska–Lincoln

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Where cancers are caught, and whether anyone looks

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Target application

The chip

Analog front end

Operating modes

Analog blocks

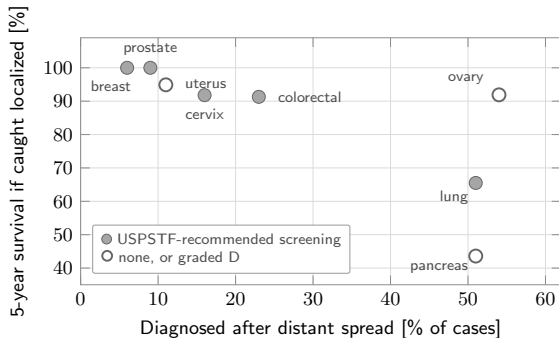
Digital

Vision

From currents to a score

References

Backup



[1] SEER 21, 2016–2022; [2, 3] USPSTF

Why ovarian: late, lethal, and unscreened

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Digital

Vision

From currents to a score

References

Backup



Site	Distant at dx	Deaths/100 cases	Screening (USPSTF)
Ovary	54 %	59	D
Pancreas	51 %	78	D
Lung	51 %	54	B, LDCT
Colorectal	23 %	35	A
Cervix	16 %	31	A
Uterus	11 %	21	none
Prostate	9 %	11	C, PSA
Breast	6 %	13	B

USPSTF: A/B recommended; C offered selectively; D recommended *against*; none: no statement [7].

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

One serum aliquot, four sites, one score

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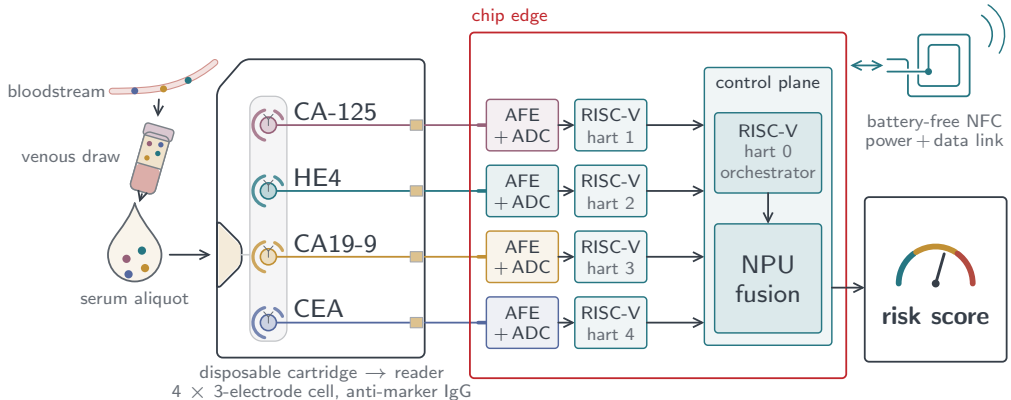
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References

Backup



Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.

No single marker is a test; four fused are [5]

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The chip

Analog front end

Operating modes

Analog blocks

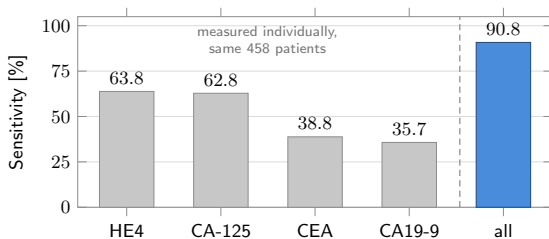
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References

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Marker	Molecule	Cut-off
CA-125	MUC16 mucin epitope [8]	35 U/mL
HE4	WFDC2 glycoprotein [6]	70 pmol/L
CA19-9	sialyl-Lewis ^a antigen	37 U/mL
CEA	CEACAM5 glycoprotein	5 ng/mL

Each marker covers a subtype the others miss. The decision is a learned combination, so the chip is four channels plus the processor that fuses them.

Block diagram

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The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

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References

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Four channel tiles own four AFE sites; every master reaches ROM, RAM, NPU and peripherals only through the arbiter.

Physical implementation

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The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

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a score

References

Backup

ARTWORK PENDING



**ARTWORK
PENDING**



Grounded-WE two-amplifier potentiostat

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The chip

Analog front end

Operating modes

Analog blocks

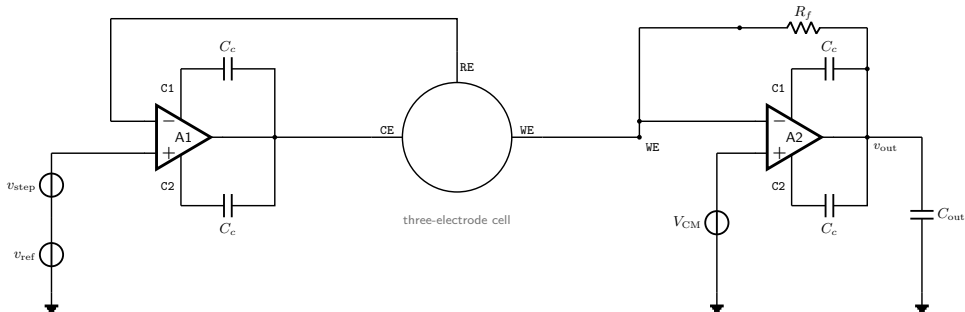
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A1 drives CE with RE in feedback; A2 holds WE at v_{cm} through R_f . $v_{out} = v_{cm} - I_{we}R_f$, both polarities.

Analog top level

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The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

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a score

References

Backup

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Virtuoso export of the analog top level: four channels, shared bias, test bus

The electrode model, in Verilog-A

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The chip

Analog front end

Operating modes

Analog blocks

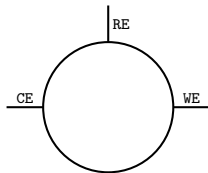
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R_u , R_{ce} , C_{dl} lumped; the faradaic branch Z_f is the Verilog-A element.

i_f faradaic current
 η overpotential, $E - E^{0'}$
 f F/RT
 $C_{O,R}(0, t)$ surface conc. of O, R
 C^* bulk conc.
 δ Nernst layer

Kinetics $i_f = nFAk^0 [C_R(0, t) e^{(1-\alpha)nf\eta} - C_O(0, t) e^{-\alpha nf\eta}]$
 Transport $\partial C / \partial t = D \partial^2 C / \partial x^2$, finite-volume, 28 expanding slabs per species
 Boundaries flux = $\pm i_f / nFA$ at the surface; $C(\delta, t) = C^*$ at the Nernst layer

Parameter	Symbol	Default	Unit	Quantity
E0	$E^{0'}$	0.0	V	formal potential of the couple
nel	n	1	—	electrons transferred
alpha	α	0.5	—	transfer coefficient
k0	k^0	0.1	cm s^{-1}	standard heterogeneous rate constant
Dif	D	5×10^{-6}	$\text{cm}^2 \text{s}^{-1}$	diffusion coefficient, both species
Area	A	0.01	cm^2	electrode area
Cox	C_O^*	1×10^{-6}	mol cm^{-3}	bulk concentration, oxidised species
Cred	C_R^*	0	mol cm^{-3}	bulk concentration, reduced species
Temp	T	298.15	K	interface temperature
expmax	—	40	—	exponent clamp on the kinetic terms
gpar	—	1×10^{-15}	S	DC shunt across the terminals

Equations: Bard–Faulkner [13]. Discretisation: Britz–Strutwolf, Feldberg [17, 18]. Precedent for Verilog-A electrode co-simulation: Jog et al. [19].

Emulating the four markers

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The chip

Analog front-end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

Marker	Reporter	$E^{0'}$ (V)	Peak (nA)	C_R^* (μ M)	Cut-off
CA-125	methylene blue	-0.15	856	279	35 U/mL [5, 8]
HE4	ferrocene	+0.10	1311	394	70 pmol/L [6]
CA19-9	TMB / HRP [20]	+0.25	1659	458	37 U/mL [5]
CEA	anthraquinone	+0.40	939	303	5 ng/mL [5, 14]

$E^{0'}$ reporter formal potential, i.e. peak position; chosen in the -0.6 to $+0.4$ V window [22, 23], 150–250 mV apart, not measured per reporter

Peak inside 50 nA to 5 μ A on screen-printed electrodes [14, 15, 21]; C_R^* fitted to reach it

k^0 0.1 cm/s, reversible; nothing published on carbon for these markers

Shared $\alpha = 0.5$, $n = 1$, $D = 5 \times 10^{-6}$ cm²/s, $A = 0.01$ cm², 37 °C, $R_f = 560$ k Ω

Representative values, not marker-measured: these slides demonstrate the channel, not an assay.

Channel range, assay range, and where the panel was placed

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The chip

Analog front end

Operating modes

Analog blocks

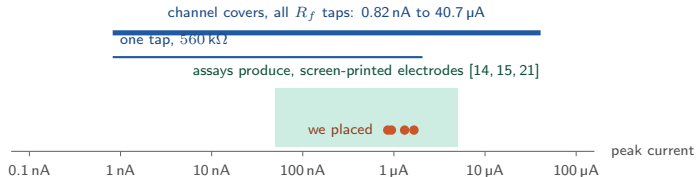
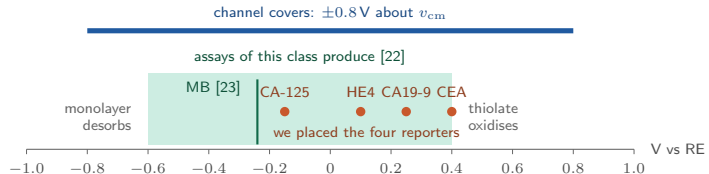
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References

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Potentials: spaced 150–250 mV for one-sweep resolution inside the reporter window, not per-reporter values. Peak heights: inside the published envelope and one tap's full scale, then C_R^* fitted. Silicon is unaffected: the channel covers the assay band with 0.2–0.4 V and a decade of current to spare, so moving the dots changes what the simulation shows, not what the chip can measure.

Exercising the cell: differential-pulse voltammetry

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The chip

Analog front end

Operating modes

Analog blocks

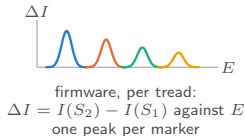
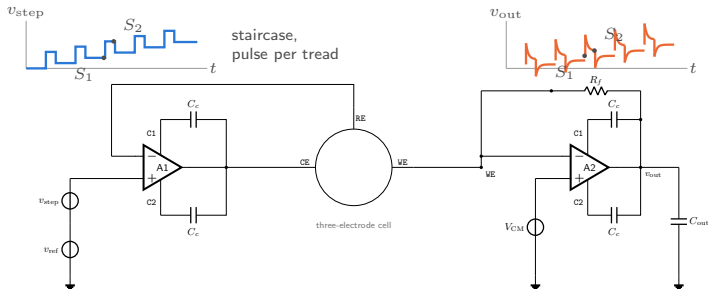
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Excite

DAC: staircase with a pulse on every tread, applied at RE by A1

Sense

$$v_{out} = v_{cm} - i_{WE} R_f$$

Sample

SAR at S_1 (before the pulse) and S_2 (end of the pulse), once charging has decayed

Resolve

firmware: $\Delta I = I(S_2) - I(S_1)$ vs E ; peak position: marker, peak height: concentration

Four markers resolved in one differential-pulse sweep

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The chip

Analog front end

Operating modes

Analog blocks

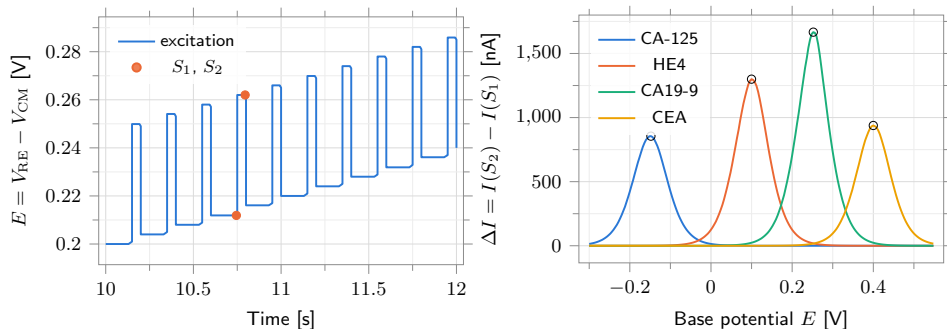
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Backup



Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

Every clinical cut-off lands inside the channel window

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The chip

Analog front end

Operating modes

Analog blocks

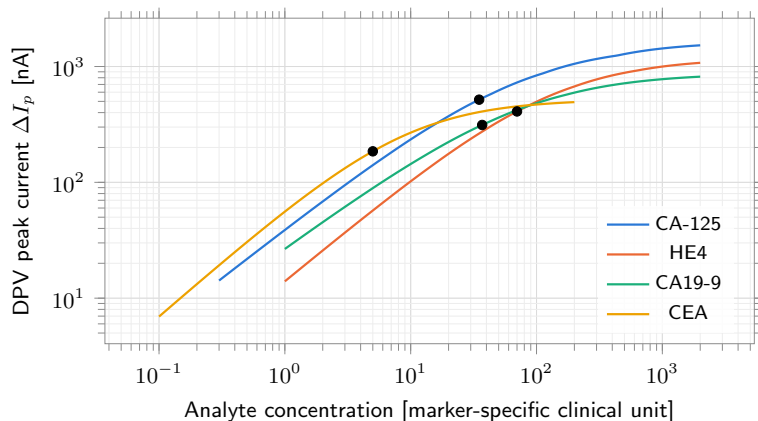
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References

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CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one R_f setting.

Cyclic voltammetry through the assembled channel

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Target
application

The chip

Analog front end

Operating modes

Analog blocks

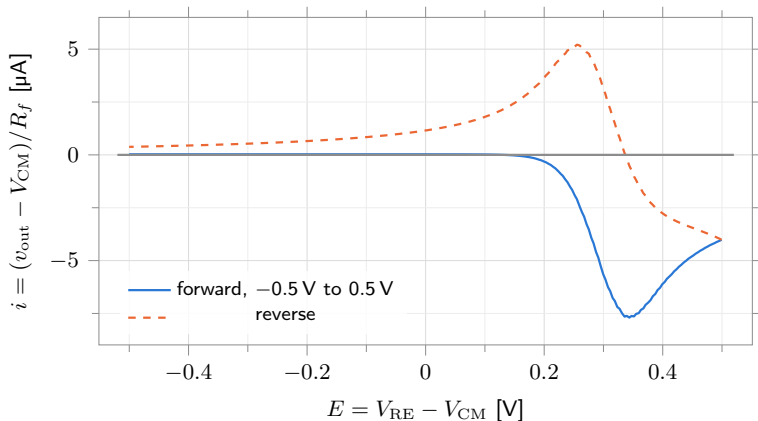
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a score

References

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Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

Why impedance: for electrodes that stay in

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Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

cartridge



single-shot sandwich immunoassay, incubate and read once

wearable

continuous glucose monitors, 10–15 days [27]
wound patch [28]; aptamer sensor, one week in vivo [29]



implant

Eversense implanted glucose monitor, 180–365 days [27]



1 min

1 h

1 day

1 week

1 month

1 year

time the electrode spends in the sample

EIS gives
Firmware

each site's $R_{ct} \parallel C_{dl}$ arc, trended over the electrode's life, from the same channel
re-ranges, recalibrates, or retires the site

Drift is systematic: averaging removes noise, not bias [25].

How impedance is measured: the same loop, no extra hardware

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Target application

The chip

Analog front end

Operating modes

Analog blocks

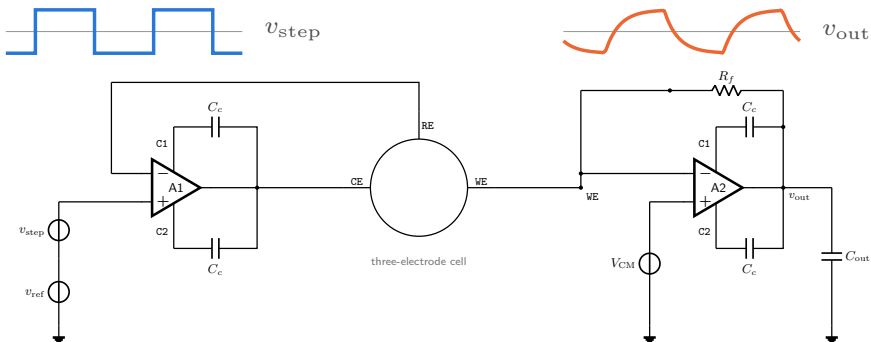
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References

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Excite
Sense
Digitise
Solve

DAC: a small two-code square about the DC setpoint v_{ref}
 v_P across the cell; $i_{WE} = -v_{out}/R_f$
the SAR records both; firmware I/Q demodulation
 $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$, or a ratio to Z_{cal}

Magnitude and phase from codes: I/Q demodulation

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Target
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

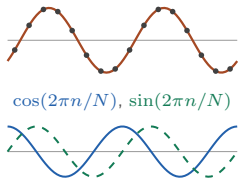
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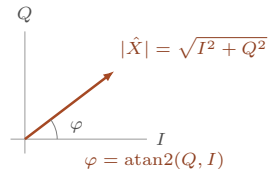
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record c_n , N per period, M periods



$$I = \frac{2}{MN} \sum c_n \cos \frac{2\pi n}{N}$$
$$Q = \frac{2}{MN} \sum c_n \sin \frac{2\pi n}{N}$$



Records v_P and v_{out}
Divide $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$
Ratio against Z_{cal} : R_f drops out

The four loads behind the EIS plots

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Target application

The chip

Analog front end

Operating modes

Analog blocks

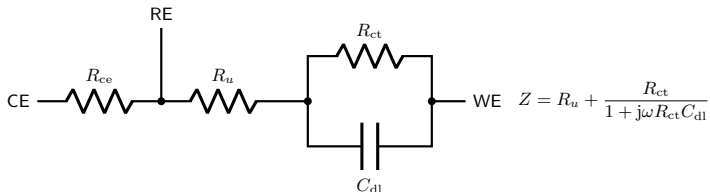
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References

Backup



Load	R_{ct}	C_{dl}	Gain code	Stands for
resistor	–	–	$N = 0$	an 11 k Ω resistor in place of the cell: no interface, so the trace is the channel's own error
fouled	10 k Ω	10 μ F	$N = 0$	a degraded electrode: adsorbed protein and biofilm open leakage paths and enlarge the interface
typical	100 k Ω	1 μ F	$N = 20$	an electrode in use
fresh	1 M Ω	100 nF	$N = 63$	a new, clean surface

$R_u = R_{ce} = 1 \text{ k}\Omega$ throughout. Fouling: R_{ct} falls and C_{dl} rises through an electrode's life; tracking that drift is what EIS is for here.

Impedance spectroscopy on the same channel

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Target application

The chip

Analog front end

Operating modes

Analog blocks

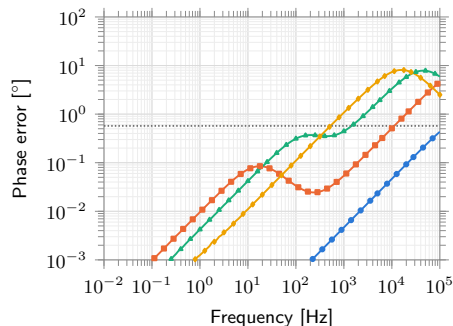
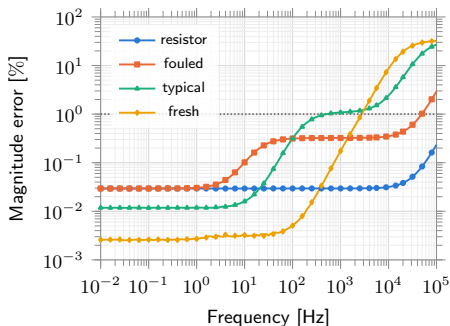
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From currents to a score

References

Backup



Accuracy $\pm 1\%$, $\pm 0.6^\circ$ to 5 kHz; $\pm 5\%$ to 27 kHz. Limit: A1's loop bandwidth
 Range 246 Ω to 12.1 M Ω . Limits: TIA compliance below, one LSB of output above; 320 R_f codes in between

The excitation amplitude never enters: Z is the ratio of two records through one converter, so DAC gain and droop cancel.

What sets the EIS error, and what averaging buys

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The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

$$\varepsilon_{|Z|} \approx \underbrace{\frac{R_f/|Z|}{A_{\text{eff}}}}_{\text{systematic}} + \underbrace{\frac{i_n |Z|}{V_1} \sqrt{\frac{2}{NM}}}_{\text{random}}$$

$R_f/|Z|$ feedback resistor over the load: the gain code chosen

A_{eff} effective loop gain, ≈ 76 dB measured (0.66 % at $R_f/R = 40$); $A_{\text{OL}} \approx 83$ dB

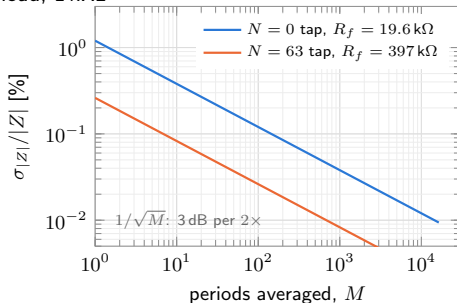
i_n front-end current noise per sample: 273 nA at $N=0$, 61 nA at $N=63$

V_1 excitation fundamental, $\frac{4}{\pi} \frac{A}{2}$

N, M samples per period, periods averaged

The systematic term is fixed for a gain code and load, and the Z_{cal} ratio removes it. The random term is what remains, and it falls as $1/\sqrt{M}$.

Random term against periods averaged, 11 k Ω load, 1 kHz



Calibrating impedance

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application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to
a score

References

Backup

$$\hat{Z} = Z_{\text{cal}} \frac{(\hat{V}_P / \hat{V}_{\text{OUT}})}{(\hat{V}_P / \hat{V}_{\text{OUT}})_{\text{cal}}} \quad \text{same gain code, same frequency}$$

	On-chip poly, per decade	Off-chip standard, at test
Cancels	R_f , its pole, DAC, LSB, loop-gain floor	the same
Known to	$\sim 20\%$ chip to chip	$< 0.1\%$
Run	any time, in the field	once, stored in flash
Good for	trends: fouling is a change in one electrode's Z	absolute ohms

The poly need not be accurate, only stable. One stored number pins the scale; the on-chip ratio carries it through temperature and time.

The class-AB amplifier: one cell, both roles

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Target
application

The chip

Analog front end

Operating modes

Analog blocks

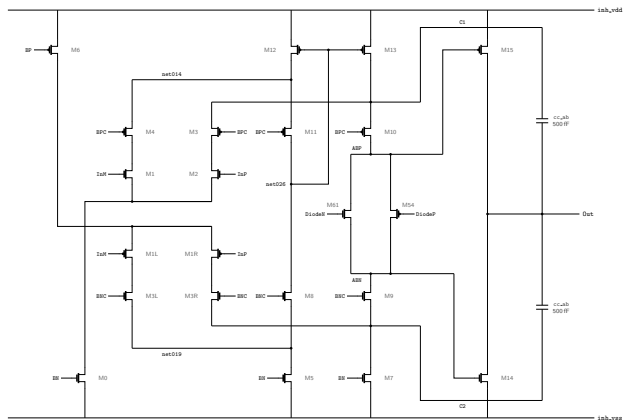
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From currents to
a score

References

Backup



complementary
input pairs and
their cascodes

diode-connected
summing branch

class-AB
drive

output

testbench
compensation

The amplifier across corners, with mismatch σ

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Target
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to
a score

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss	σ (MC)
Open-loop gain	dB	82.77	79.88	83.14	83.37	84.94	1.06
Phase margin	deg	73.52	74.70	74.04	72.86	73.34	1.42
Dominant pole	kHz	1.592	2.286	1.538	1.494	1.225	0.276
Gain-bandwidth product	MHz	22.78	23.19	22.89	22.94	22.59	3.57
Input offset voltage	mV	0.622	0.649	0.624	0.569	0.456	5.201

Bipolar accuracy through zero

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Target application

The chip

Analog front end

Operating modes

Analog blocks

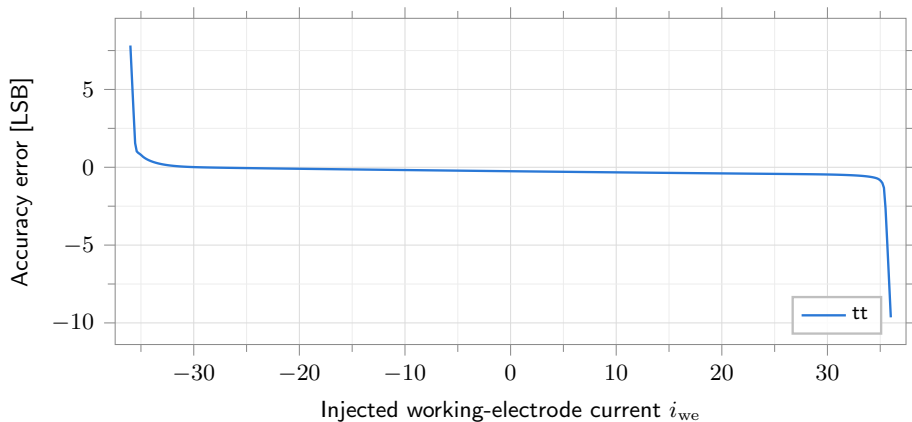
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Programmable feedback resistor: one string, 320 codes

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The chip

Analog front end

Operating modes

Analog blocks

Digital

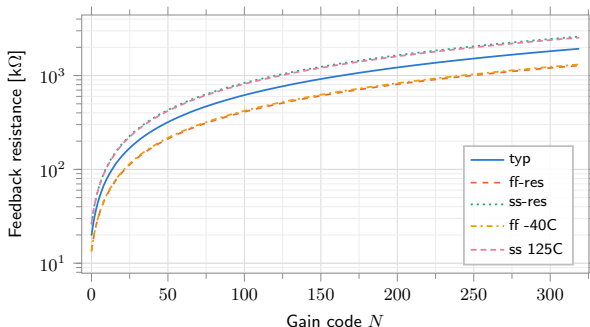
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References

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- $R_f = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$; full scale = $0.80 \text{ V}/R_f$, LSB = $1.59 \text{ mV}/R_f$
- **Max $\pm 40.7 \mu\text{A}$, min full scale $\pm 0.41 \mu\text{A}$, resolution 0.82 nA**
- Noise floor $i_n^2 = e_n^2 (1 + R_f/|Z_{el}|)^2/R_f^2 + 4kT/R_f$: 341 pA alone, 3.6–21 nA into an electrode (0.1–100 Hz, 560 k Ω)



N	R_f	Full scale	LSB
0	19.6 k Ω	$\pm 40.7 \mu\text{A}$	81 nA
63	397 k Ω	$\pm 2.02 \mu\text{A}$	4.0 nA
319	1.93 M Ω	$\pm 0.41 \mu\text{A}$	0.82 nA

Mismatch σ	σ/typ	
$R(N=0)$	71 Ω	0.36 %
$R(N=319)$	0.7 k Ω	0.04 %
Seam 63 $\rightarrow$ 64	504 Ω	6.8 %

10-bit SAR ADC, extracted netlist

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Target
application

The chip

Analog front end

Operating modes

Analog blocks

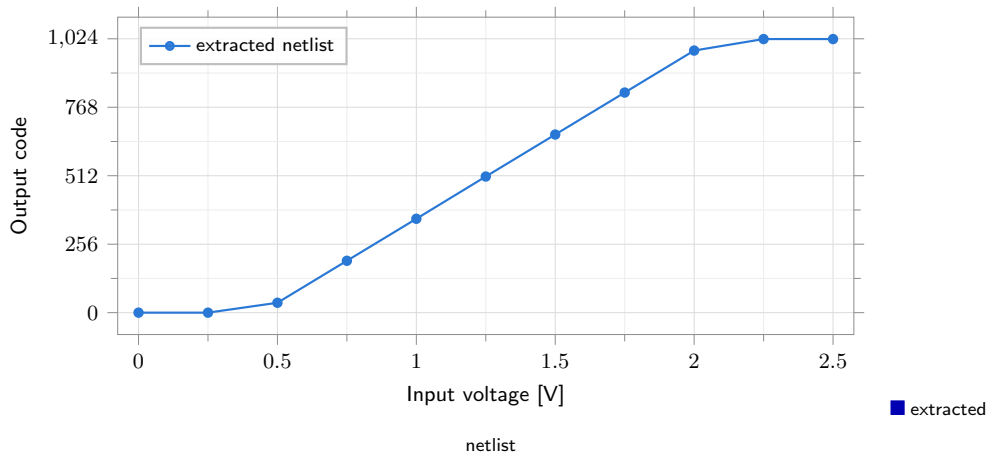
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23 clocks per conversion, 870 kSa/s, 50 pJ. The capacitor array exists only in layout, so this is the one block characterised extracted.

SAR ADC across corners

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The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss
Fitted slope	codes/V	628.59	628.59	628.40	628.73	628.13
Effective LSB	mV	1.5909	1.5909	1.5914	1.5905	1.5920
Input at code 0	V	0.4408	0.4408	0.4407	0.4408	0.4399
Usable span	V	1.6274	1.6274	1.6280	1.6271	1.6286
Worst residual	LSB	0.495	0.495	0.490	0.477	0.484
RMS residual	LSB	0.294	0.294	0.277	0.288	0.302
Analog supply current	μ A	6.36	4.79	5.45	8.44	15.43
Energy per conversion	pJ	20.8	19.3	18.1	27.3	49.5

Reference DAC across corners, with mismatch σ

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Target
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to
a score

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss	σ (MC)
Full-scale output	V	2.499388	2.499388	2.499388	2.499388	2.499388	0.000001
LSB size	μV	610.351	610.351	610.351	610.351	610.351	0.0003
INL, most positive	LSB	0.113	0.101	0.100	0.127	0.130	0.610
INL, most negative	LSB	-0.376	-0.338	-0.390	-0.362	-0.414	0.620
DNL, most positive	LSB	0.002	0.001	0.002	0.002	0.002	1.128
DNL, most negative	LSB	-0.423	-0.374	-0.424	-0.424	-0.479	1.415

The front end, against the field

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Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

	Castalia (this work)	Myshkin [9]	Chen '19 [10]	Ghoreishizadeh '17 [11]	Lin '23 [12]
Modality	CA/CV/DPV/EIS	CA/CV	CA/CV	CA/CV	CV/EIS
Channels	4, simultaneous	1	1	1	1
Current direction	bipolar	unipolar	bipolar	bipolar	bipolar
$I_{\max}$ (μA)	± 40.7	+20	± 5	± 20	$-48/+16$
Resolution (pA rms)	341^a	56.5	41	0.47	1.7
ADC step (pA)	820^b	153	n/a	n/a	n/a
Converter	10-bit SAR per channel	12-bit dual-slope	TDC	ADC	–
EIS	firmware, $\pm 1\%$ to 5 kHz	–	–	–	yes
Power (mW)	3.68^c	3.0	0.016	9.3	1.0
Process (nm)	65	65	180	350	180
On-chip processor	5 RISC-V harts	1 RISC-V hart	none	none	none
On-chip ML	NPU, 4 modes	NPU, 1 mode	none	none	none

^aInput-referred noise, 0.1–100 Hz, ideal R_f , no electrode.

^bOne converter code at the maximum-gain tap; 81 nA at the lowest.

^cLow-power mode: hart 0 awake, tiles gated, AFE off. One channel measuring adds 1.76 mW (tile) + 1.32 mW (AFE).

Analog power: one channel

Castalia

Maxx Seminario

Target
application

The chip

Analog front end

Operating modes

Analog blocks

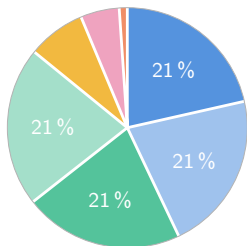
Digital

Vision

From currents to
a score

References

Backup



- A1, control amplifier, 244 μW , 21 %
- A2, transimpedance amplifier, 244 μW , 21 %
- ADC input buffer, 244 μW , 21 %
- debug buffer, 244 μW , 21 %
- two R-2R ladders, 89 μW , 8 %
- local bias generator, 59 μW , 5 %
- SAR converter, 12 μW , 1 %

Power

One channel	1.32 mW
Four channels	5.3 mW

Castalia at a glance

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Digital

Harts	5 (1 orchestrator, 4 channel tiles)
ISA	rv32imac_zba_zbb_zbs_zbc
Clock	24 MHz rated, closed at 25 MHz
Measured CPI	1.260 aggregate, nine benchmarks
Private TCM	8 KiB per hart, 1-cycle
Shared RAM	80 KiB behind a round-robin arbiter
Synchronisation	16 hardware mutexes, LR/SC, AMOs
Interrupts	claim/complete, any vector to any hart
Accelerator	NPU: MLP, conv-1D, XNOR, GEMM
Power domains	4 MTCMOS, one per channel tile
Debug	JTAG, RISC-V Debug Module
Field boot	NFC ISO 14443A harvested boot

Analog and physical

Measurement sites	4 identical, simultaneous
Topology	bipolar grounded-WE potentiostat
Electrode pads	12 (CE/RE/WE $\times$ 4)
Converter	10-bit SAR per channel, extracted
Reference DAC	12-bit R-2R per channel
Feedback resistor	$18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$, 320 codes
Full scale	$\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$
EIS	firmware mode of the same channels
Analog supply	2.5 V , $v_{\text{cm}} = 1.25\text{ V}$
Process	65 nm CMOS, 1P8M, HVT cells
Die / package	$3.0 \times 3.0\text{ mm}$ / LQFP-100
Digital power	12.8 mW core post-route, SS, 25 MHz

Digital numbers from the hardened flow; analog numbers schematic-level, except the SAR, which is extracted.

Digital, as closed

Castalia

Maxx Seminario

Target
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to
a score

References

Backup

Digital, post-route

Internal	8.69 mW
Switching	9.99 mW
Leakage	4.70 mW

Core logic	12.8 mW
Pad ring*	10.5 mW
Total	23.4 mW

One channel tile

Total	1.761 mW
Gated	0

Measured CPI, timed kernels

Benchmark	Instr.	CPI
spmv	802 447	1.143
multiply	20 892	1.163
vvadd	2 408	1.374
median	4 015	1.398
qsort	123 205	1.398
rsort	146 548	1.412
dhrystone	200 523	1.504
towers	3 747	1.682
memcpy	11 017	1.726

Aggregate	1 314 802	1.260
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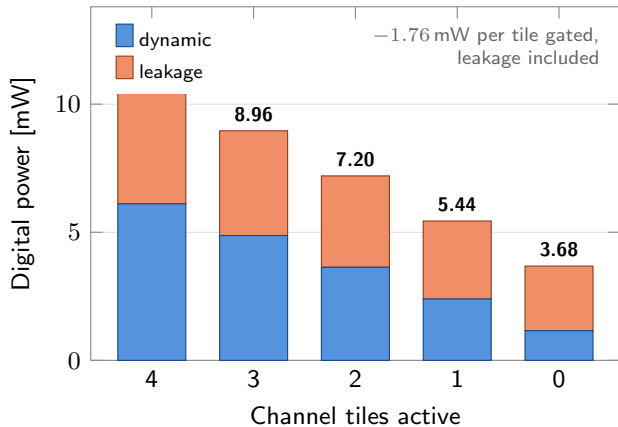
SS 0.9 V 125 °C, 25 MHz, statistical activity 0.2; d13d cut. *Pads at the statistical activity, to be optimised.

■ hardened flow, post-route

Power gating: one switched rail per channel tile

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■ hardened flow, post-route

Cold gating: the header cuts the rail, so a parked tile draws neither dynamic nor leakage power. Each tile gated removes 1.76 mW; the always-on band keeps 3.68 mW.

Vision: the same channel, without the cartridge

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

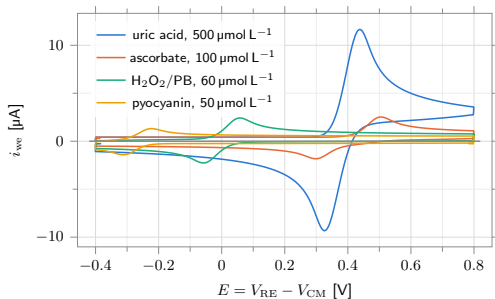
Digital

Vision

From currents to a score

References

Backup



Already on the die

4 sites	concurrent, each its own DAC, SAR, hart
EIS	electrode state without a second instrument
NFC	harvested boot, battery-free readout
MTCMOS	a parked site costs nothing
NPU	the decision stays on the patient

Left: a small-molecule wound panel, uric acid, ascorbate, H_2O_2 and pyocyanin, through the same channel.

The ovarian panel stays a serum test: CA-125 is an epitope on a 3–5 MDa mucin and does not reach interstitial fluid. The wearable is a different, reagentless panel on the same silicon; the implant is the same plus a fouling problem that EIS can watch but not solve.

The fusion runs on the NPU

Castalia

Maxx Seminario

Target
application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to
a score

References

Backup

$$s = \sigma \left(\sum_{k=1}^4 w_k \hat{c}_k + b \right)$$

Input	peak current, peak potential, baseline $\times 4$
Mode	0, dense layer, one sigmoid
Output	one risk score
Weights	trained off-chip, staged by firmware

Per site

Hart	1, ownership enforced in hardware
Potential	own 12-bit DAC
Converter	own 10-bit SAR
Power	own MTCMOS domain

The logistic form of the cleared two-marker algorithm, extended to four inputs and moved from the central laboratory onto the electrode.

References (1/5)

Castalia

Maxx Seminario

Target
application

The chip

Analys front end

Operating modes

Analys blocks

Digital

Vision

From currents to
a score

References

Backup

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References (2/5)

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

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References (3/5)

Castalia

Maxx Seminario

Target
application

The chip

Analogue front end

Operating modes

Analogue blocks

Digital

Vision

From currents to
a score

References

Backup

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References (5/5)

Castalia

Maxx Seminario

Target application

The chip

Analogue front end

Operating modes

Analogue blocks

Digital

Vision

From currents to a score

References

Backup

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Backup: electrode model validation

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

Scan rate mV s^{-1}	i_p Randles–Ševčík μA	i_p model μA	Error %	Error, $C_{\text{dl}} \rightarrow 0$ %	ΔE_p mV
25	0.951	0.952	+0.08	−0.45	58.78
50	1.345	1.349	+0.32	−0.42	59.85
100	1.902	1.915	+0.67	−0.38	60.41
200	2.690	2.721	+1.16	−0.33	62.33
400	3.804	3.875	+1.86	−0.25	64.74

Peak current against Randles–Ševčík and ΔE_p against Nicholson [13], reversible couple, $C_{\text{dl}} = 0$ column isolates the faradaic model.

Backup: USPSTF screening grades

Castalia

Maxx Seminario

Target application

The chip

Analog front end

Operating modes

Analog blocks

Digital

Vision

From currents to a score

References

Backup

Grade	Certainty of net benefit	Definition [7]	Cancer screening [2, 3]
A	high, substantial	Recommends the service. Offer or provide it.	cervix (2018), colorectal 50–75 (2021)
B	high moderate, or moderate	Recommends the service. Offer or provide it.	breast 40–74 (2024), lung LDCT 50–80 (2021), colorectal 45–49 (2021) prostate PSA 55–69 (2018)
C	at least moderate, small	Offer selectively, by professional judgment and patient preference.	
D	no net benefit, or harms outweigh	Recommends <i>against</i> the service. Discourage its use.	ovary (2018) , pancreas (2019)
I	cannot be determined	Evidence insufficient, poor, or conflicting.	—
none	—	No USPSTF statement exists for the condition.	uterus

Ovarian D is a verdict on a test that was tried, CA-125 plus transvaginal ultrasound [2, 4], not on the absence of one.